



IDX G9 GEOGRAPHY H STUDY GUIDE ISSUE S2

FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

—**Salinity:** concentration of dissolved salt in a solution. Seawater's average is 35 parts per thousand

—**Brackish water:** water that has a salinity of less than 35 parts per thousand

—**Brine:** a water that has a salinity of more than 35 parts per thousand

—**Mixing zone:** 2% of the oceanic mass. Solutes are blended here

—**Thermocline transition zone:** 18% of the oceanic mass. in the top two layers, temperature, salinity, and oxygen decrease. CO₂ increases here as you get lower.

—**Deep cold zone:** bottom 80% of the ocean. temperature and salinity values are uniform. Coldest water is at the bottom

Factors driving coastal systems

—Solar energy input drives the atmosphere and hydrosphere. Conversion of insolation to KE produces prevailing winds, weather systems, and climate.

—Winds generate ocean currents and waves

—Climatic regimes, which result from insolation and moisture, also influence coastal processes

—Coastal rock hardness. Hard rock is harder to erode than soft rock.

—Human activities also play a part

—**Littoral zone:** coastal area and shallow offshore environment

—**Shoreline:** line of contact between sea and land. shifts with tides etc.

—**Intertidal zone:** foreshore

—**Mean sea level:** value based on tidal levels recorded hourly over many years

—**Tides:** twice-daily oscillations in sea level

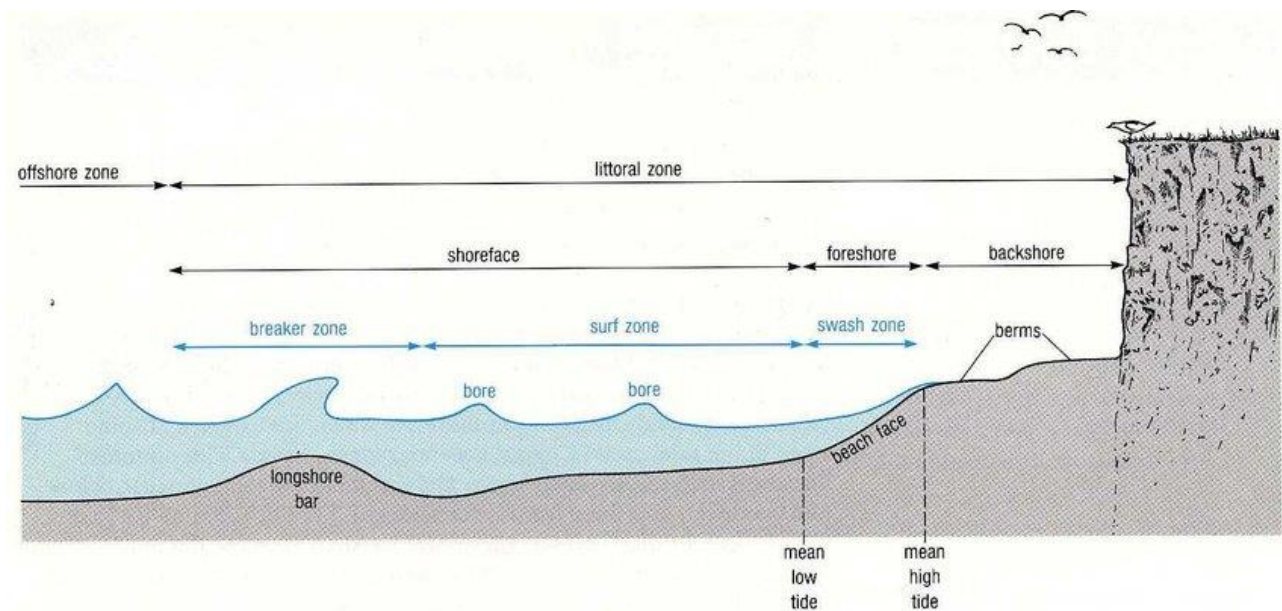
—**Tidal bulges:** fluid stretching caused by the moon's gravitational pull

—**Flood tides:** high tide

—**Ebb tides:** low tide

—**Spring tide:** when the sun and moon are on the same side of earth. highest ocean levels

—**Neap tide:** when the sun and moon are at right angles to the earth. high low tides and low high tides.



—**Waves:** undulations of water

—**Wave trains:** groups waves move in

—**Swells:** smooth, rounded waves.

- Wave energy:** instead of the water itself moving, energy is transported through water
- Waves of transition:** simple cyclic motions water moves in
- Waves of translation:** when both energy and water moves towards shore
- Rip current:** when backwash caused by breakers flows into the ocean from the beach in a concentrated column
- Interference:** how much waves mess with each other
 - if waves are in phase, height is amplified. If they are not, they become weaker

Possible Factors that Change Sea Level

- Tides and waves processes
- Long-term changes in global temperature and sea level

Zones of Wave Influence:

- breaker zone, surf zone, swash zone ($\approx$ foreshore)

Zones of Tide Influence:

- backshore, foreshore, nearshore

The Sediment Profile:

- berms, beach face, longshore bar

What Forces Drive Coastal Systems?

- Internal Forces:
 - tectonic movement
 - volcanic eruption & earthquake
 - vegetation stabilization (e.g. mangroves, salt marshes, dune grasses trapping sediment)
 - wide and long shelf (it reduces wave energy, and allows coast to build seaward)

--lithology (harder or softer rock on coasts) (e.g. limestone cliffs erode fast) (e.g. rocky/sandy/muddy/mangrove/coral-reef coasts)

--External Forces:

--atmospheric (wind, climate)

--ocean (waves/storms, tides, eustatic rise, currents)

--gravitational attraction (moon + earth + sun = tides)

--land (rivers) (size and type of sediment depends on river and distance of transport)

Types of Waves:

--wind waves (caused by wind)

--swell waves (caused by wind and then inertia)

--tsunami (caused by sudden and sharp motions in the seafloor – earthquakes, submarine landslides, or eruptions of undersea volcanoes)

--tidal waves (caused by gravitational force of the moon and sun)

--breakers/breaking waves (caused by friction of seabed)

Wind Waves:

--generated by winds in a stormy area at open ocean (known as fetch area)

--height of waves depends on wind velocity, duration, and fetch length

--wave height ranges from mm to over 100 ft

--KE and PE of waves increase in the fetch area

Swell Waves:

--travelled outside of fetch area → not affected by wind anymore but keeps moving due to inertia

--becomes smooth, round-shaped, long-crested because of the friction between air and water

--uniformly symmetrical waves, little loss of energy

Tsunamis:

- triggered by sudden and sharp motions in the seafloor, hard to detect or forecast
- 3 in. to 1 m tall in open sea, length can reach over 100 km
- travel at a great speed (600-800 km/h)
- get much taller and shorter in length when approaching shore

Breaking Waves:**--Spilling Breakers:**

- forms on flat/gentle seafloor → gentle waves
- wave energy slowly dissipates

--Plunging Breakers:

- steeper seafloor or sudden depth changes
- crest becomes vertical and then curls over; releases most of its energy at once

--Surging Breakers:

- when low steepness waves meet very steep beach
- crests remain relatively unbroken as waves slide up the beach

Constructive Waves:

- deposits more sediment than it erodes; stronger swash

Destructive Waves:

- erodes more sediment than it deposits; stronger backwash

Oceans and Coastal systems

- Wind waves: caused by winds
- Swell waves: caused by winds, then inertia

- Tsunami (harbor waves): caused by earthquakes/undersea volcano eruptions. Hard to detect, because they travel very quickly and have a very low height in the open ocean, while having a wavelength of over 100km.
- Tidal waves: caused by tides
- Breakers: caused by friction
- Wind wave trains are generated by winds in a stormy area at the open ocean
- Height is determined by wind velocity, duration, fetch length. Can range from millimeters to over 100 ft
- KE and PE of waves increase in the fetch area

Formula for wave base: $d = \frac{L}{2}$

Deep water waves: $d \geq \frac{L}{2}$

Transitional water waves: $\frac{L}{20} < d < \frac{L}{2}$

Shallow water waves: $d \leq \frac{L}{20}$

- In general, waves curl over and break when either $d \leq 1.3h$ or $\frac{H}{L} \geq \frac{1}{7}$

Kinds of breakers

- Spilling breakers: flat, gentle seafloor = gentle waves. Waves get steeper until the crest becomes unstable, leading to whitewater (foam) covering the leading face of the wave. Its energy is slowly dissipated in the whitewater.
- Plunging breakers: steeper sea floor/sudden depth changes, like a sandbar. The crest becomes vertical, then curls over, then drops onto the trough of the wave. Releases most of its energy at once.

- Surging breakers: when low steepness waves meet at a very steep beach. The wave crest disappears very quickly as a result of the rapid movement of the wave base. Crests remain unbroken. Little foam.

Coastal system outputs

Destructive waves	Constructive waves
Stormy conditions	Calm weather
High wave energy	Low wave energy
Backwash is stronger than swash	Swash is stronger than backwash
Short wavelength, high height & frequency	Longer wavelength, low height & frequency
Longer fetch	Shorter fetch

Coastal erosion

- Erosional coastlines are rugged, of high relief, and tectonically active
- Sea cliffs: formed by undercutting action of the sea. Can become notched and will collapse and retreat eventually
- Sea caves: cave that opens to the sea
- Sea arches: arch caused by erosion
- Sea stacks: caused by collapsing arches, leaving isolated towers in the water
- Wave-cut platform/Wave-cut terrace: caused by wave action cutting a horizontal bench in the tidal zone, extending from the foot of a sea cliff out into the sea. Show up in places with lots of tectonic activity.

Coastal deposition

- Barrier spits: material deposited in a long ridge extending out from a coast, sometimes partially crossing and blocking the mouth of a bay

- Bay barrier/baymouth bar: when a spit completely cuts off a bay from the ocean
- Lagoon: shallow saltwater body that is cut off from the ocean
- Tombolo: when sediment deposits connect the shoreline with an offshore island or sea stack

Beach Protection

- Breakwater: traps sand, parallel to shore
- Jetty: protects entrance of river into sea from deposition
- Groin: traps sand (deposition on one side, erosion on the other)

Coral Reefs:

Corals:

- 30° N ~ 30° S
- $18^{\circ}\text{C} \sim 29^{\circ}\text{C}$
- 10-55m deep water
- 27‰ ~ 40‰ salinity
- Clear, transparent water with little or no sedimentation
- Vigorous water motion

Three Types of Coral Reefs:

- Fringing Reefs: grow seaward directly from the shore, and form borders along the shoreline and surrounding islands.
- Barrier Reefs: similar to fringing reefs as they also border a shoreline. However, instead of growing directly out from the shore, they are separated from land by an expanse of water. This creates a lagoon of open, often deep water between the reef and the shore.
- Atolls: a ring-shaped coral reef containing a coral rim that encircles a lagoon.